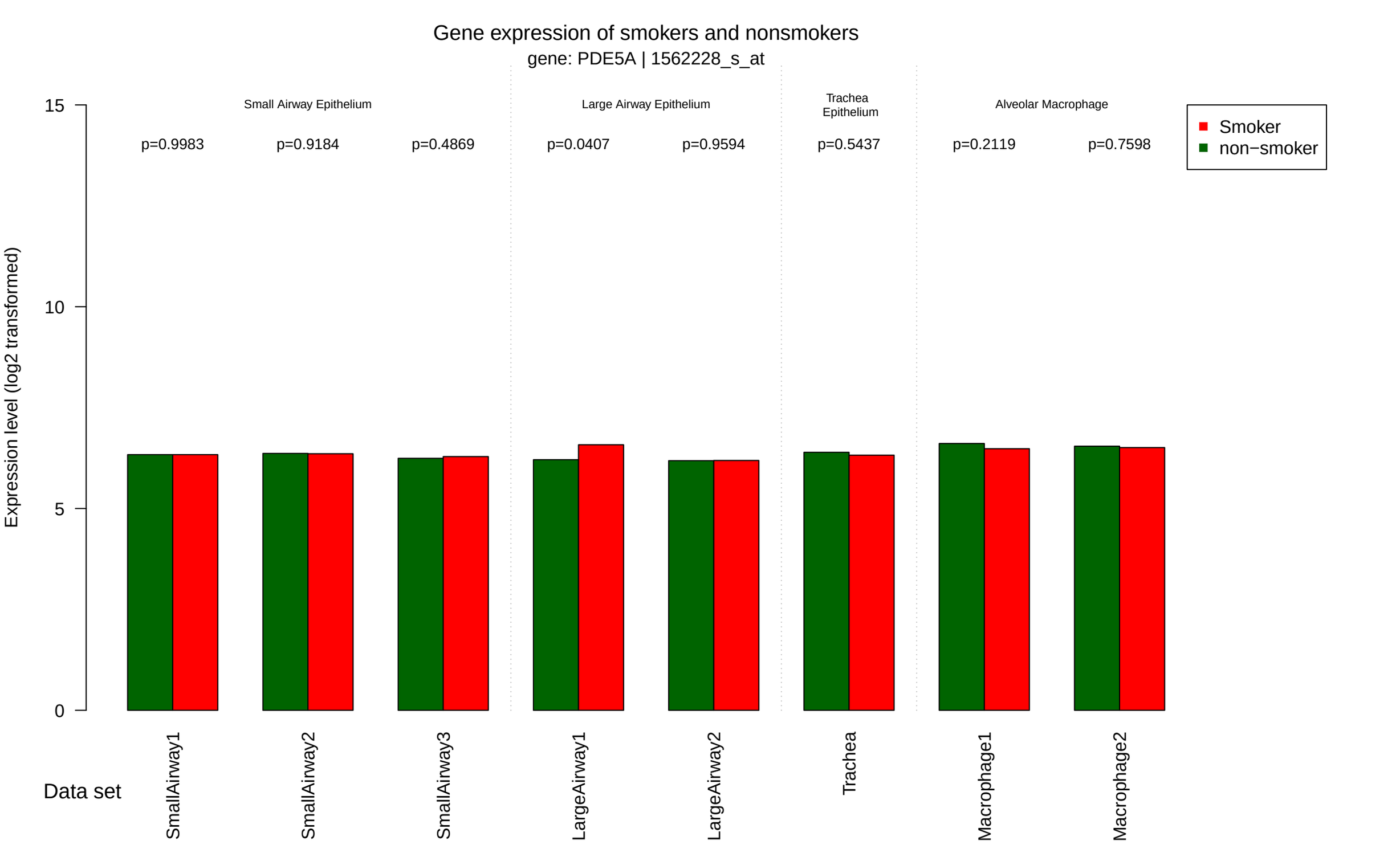
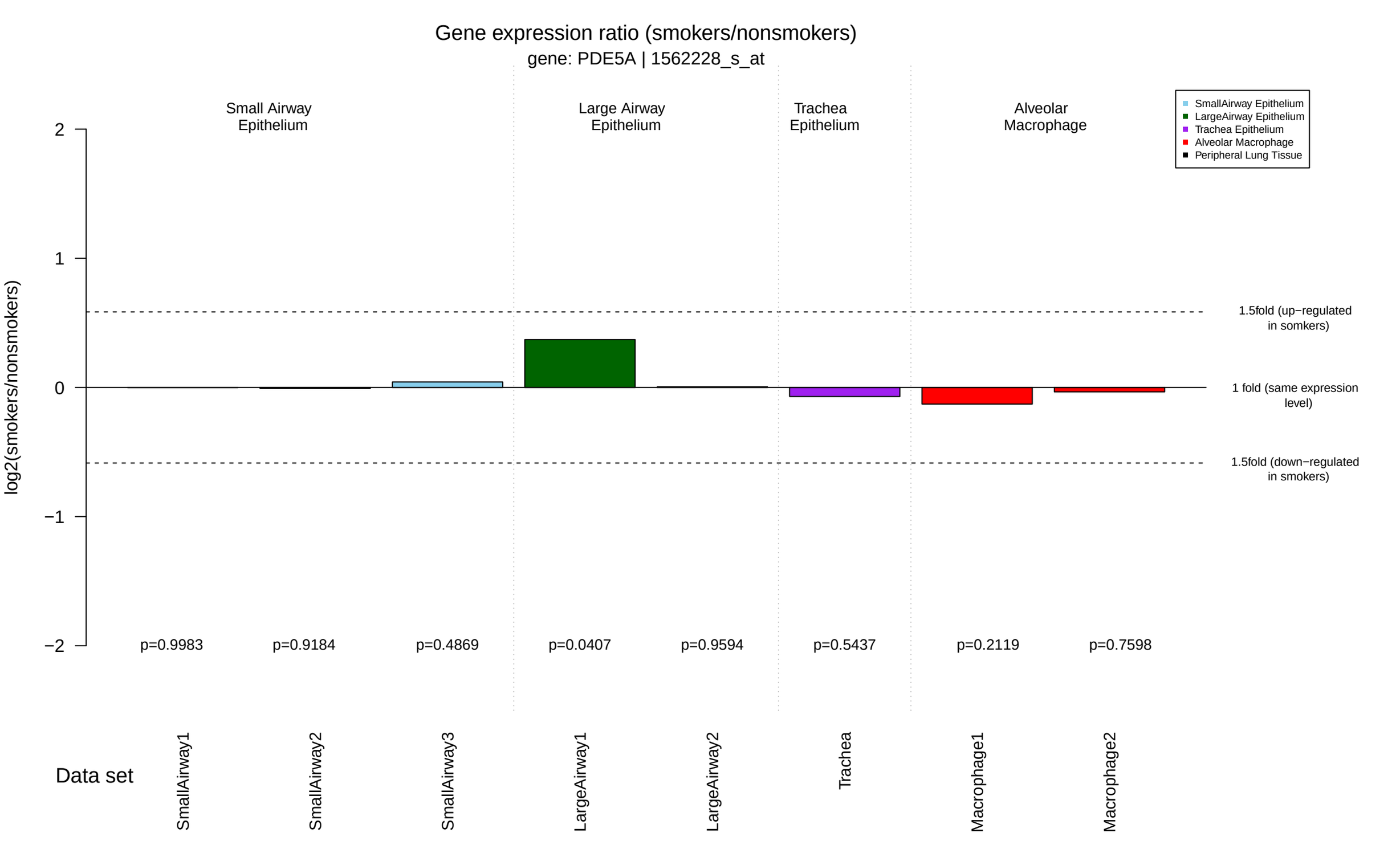
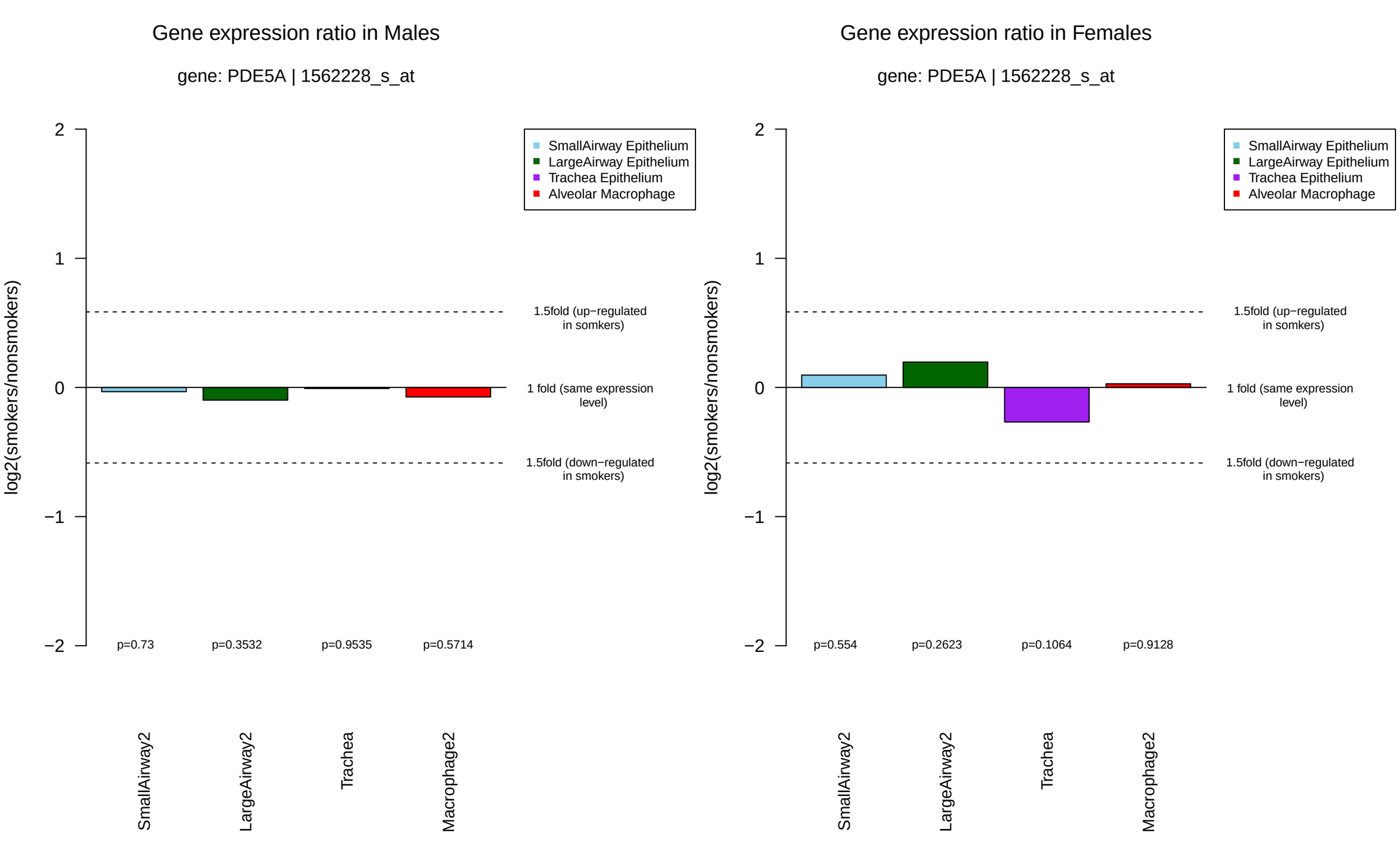
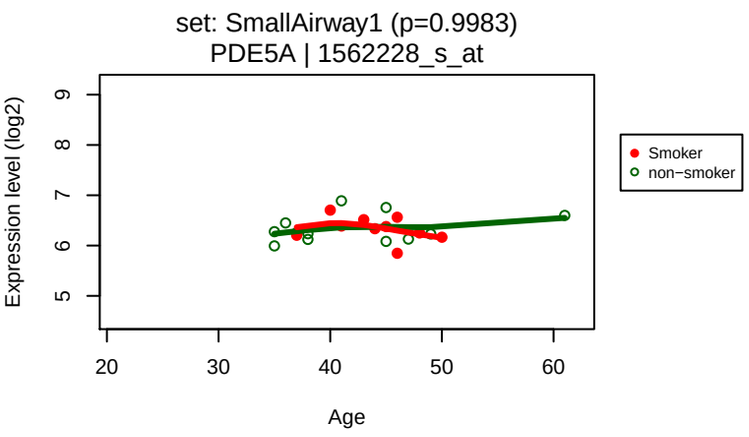
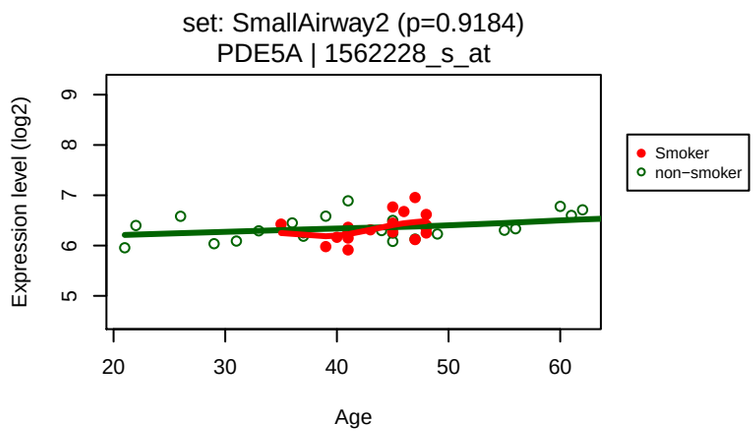
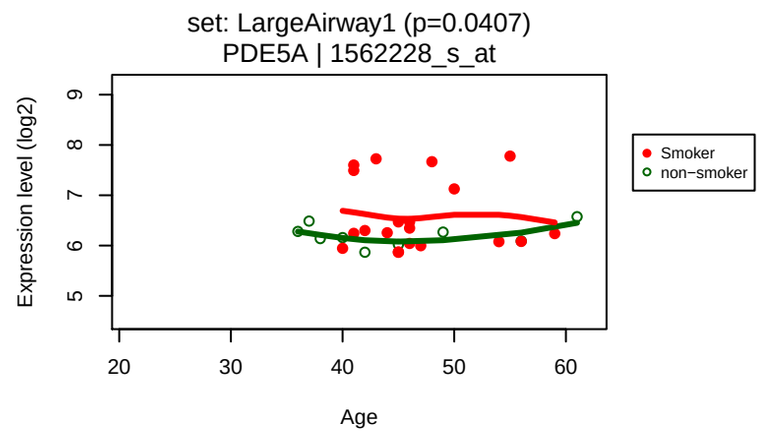

Gene expression of smokers and nonsmokers
gene: PDE5A | 1562228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9983
p=0.9184
p=0.4869
p=0.0407
p=0.9594
p=0.5437
p=0.2119
p=0.7598
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

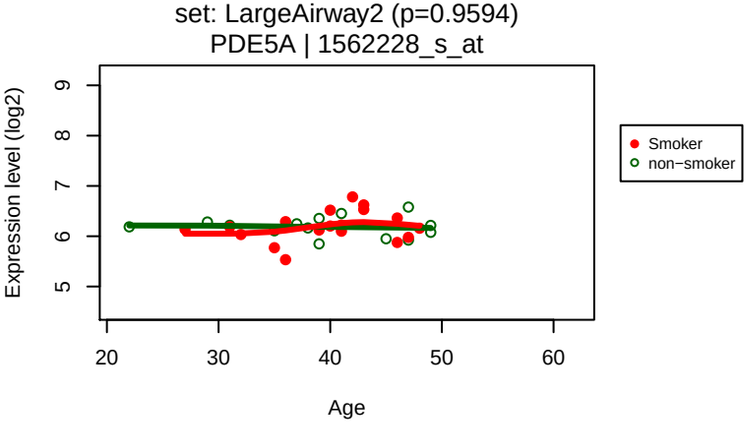

Gene expression ratio (smokers/nonsmokers)
gene: PDE5A | 1562228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9983
p=0.9184
p=0.4869
p=0.0407
p=0.9594
p=0.5437
p=0.2119
p=0.7598
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

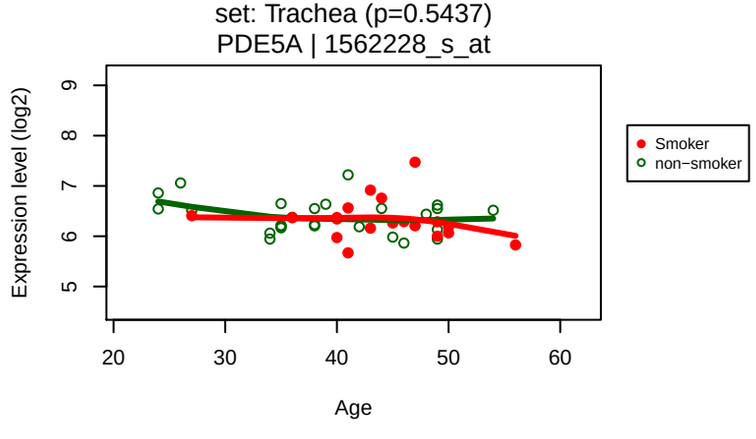

Gene expression ratio in Males
gene: PDE5A | 1562228_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.73
p=0.3532
p=0.9535
p=0.5714
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE5A | 1562228_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.554
p=0.2623
p=0.1064
p=0.9128
SmallAirway2
LargeAirway2
Trachea
Macrophage2

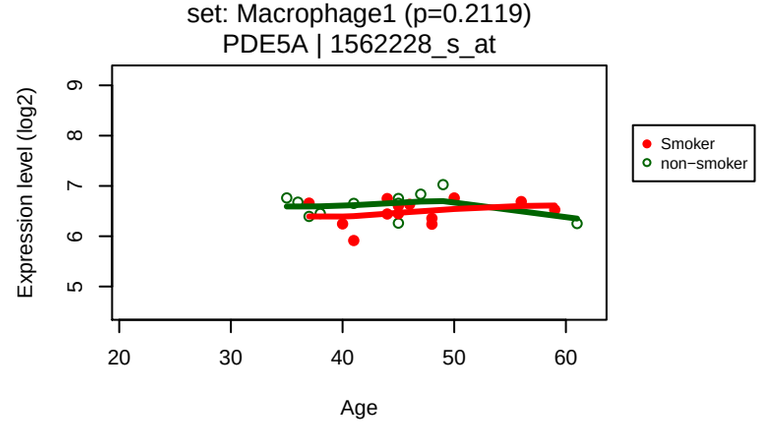

set: SmallAirway1 (p=0.9983)
PDE5A | 1562228_s_at
Expression level (log2)
Age
Smoker
non−smoker

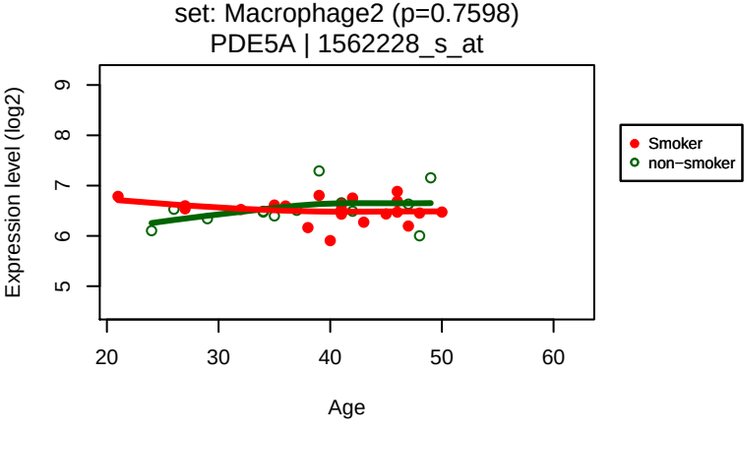

set: SmallAirway2 (p=0.9184)
PDE5A | 1562228_s_at
Expression level (log2)
Age
Smoker
non−smoker

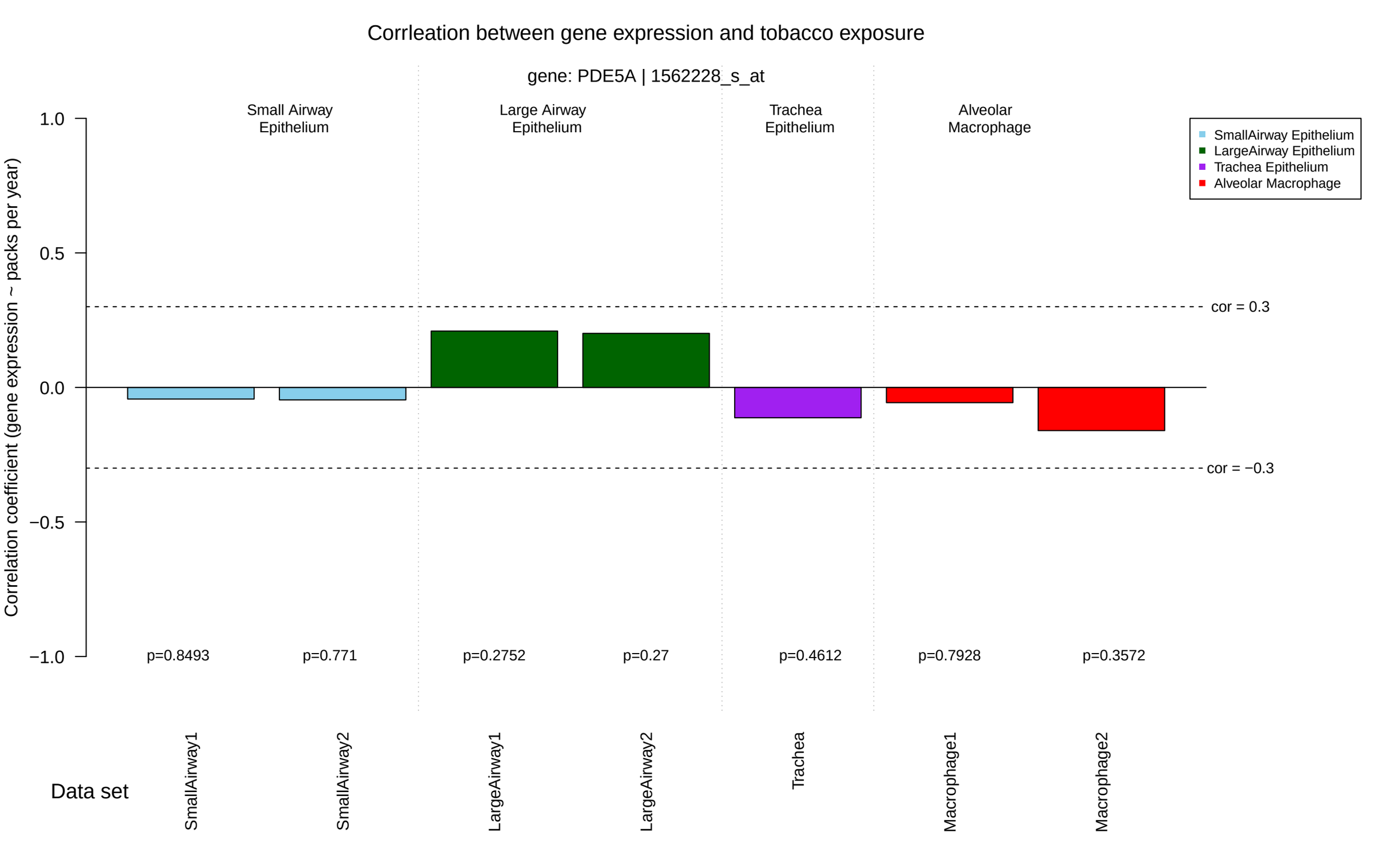

set: LargeAirway1 (p=0.0407)
PDE5A | 1562228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9594)
PDE5A | 1562228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5437)
PDE5A | 1562228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2119)
PDE5A | 1562228_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7598)
PDE5A | 1562228_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE5A | 1562228_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8493
p=0.771
p=0.2752
p=0.27
p=0.4612
p=0.7928
p=0.3572
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2